Prior-Experience-Based Vision-Language Model for Remote Sensing Image–Text Retrieval

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***Abstract*— Remote sensing (RS) image–text retrieval (RSITR) aims to retrieve relevant texts (RS images) based on the content of a given RS image (text). Existing methods are used to employing the convolutional neural network (CNN) and recurrent neural network (RNN) as encoders to learn visual and textual features for retrieval. Although feasible, the global information hidden in different data does not receive the attention it deserves. To mitigate this problem, transformers have been introduced. Nevertheless, the complexity of RS images present challenges in directly introducing Transformer-based architectures to multimodal learning in RS scenes, particularly in visual feature extraction and cross-modal interaction. In addition, the textual captions are always simpler than the complex RS images, leading to a semantic description appearing in different images. This typical false-negative (FN) sample problem increases the difficulty of RSITR tasks. To address the above limitations, we propose a new RSITR model named prior-experience-based RS vision-language (PERSVL). First, the specific visual and text encoders are used to extract features from RS images and texts. Also, a high-level feature complement (HFC) module is developed based on the self-attention mechanism (SAM) for the visual encoder to explore the complex contents from RS images fully. Second, a dual-branch multimodal fusion encoder (DBMFE) is designed to complete the cross-modal learning. It comprises a dual-branch multimodal interaction (DBMI) module and a branch fusion module. DBMI is designed to fully explore the relationships between different modalities, enriching visual and textual features. The branch fusion module integrates the cross-modal features and utilizes a classification head to generate matching scores for retrieval. Finally, a learning from prior experiences (LPEs) module is designed to reduce the influence of FN samples by analyzing the historical data produced in the model training process. Experiments are conducted on three popular datasets, and the positive results show that our PERSVL model achieves superior performance compared with previous methods. By integrating the advantages of natural language and RS images, our PERSVL can be applied in various applications, such as environmental monitoring, disaster evaluation, and urban planning. Our source codes are available at: https://github.com/TangXu-Group/Cross-modal-remote-sensing-image-and-text-retrieval-models/tree/main/PERSVL.**

***Index Terms*— Learning from prior experiences (LPEs), multiscale feature fusion, remote sensing image–text retrieval (RSITR), transformer.**

I. Introduction

With the development of Earth observation technology, the quality and quantity of remote sensing (RS) images have significantly been improved [1], [2], [3]. Such RS images can enhance human perception of the Earth and thus have various applications in environment monitoring, disaster rescue, global security, and urban planning [4], [5], [6]. However, obtaining valuable information from this diverse and large-scale data according to the users' demands is challenging. Considering natural language can well express users' requirements, the RS image–text retrieval (RSITR) technique has been introduced to the RS community to address the above obstacle in recent years.

RSITR is a bidirectional retrieval task that retrieves text (RS image) based on a query RS image (text) [7], [8], [9]. It aims to establish semantic connections between RS images and texts, which needs to explore high-level semantic information and inherent characteristics within both RS images and texts [4]. In the past several years, many RSITR methods [8], [10], [11], [12], [13] have been proposed, and they are accustomed to adopting a convolutional neural network (CNN)-based image encoder and a recurrent neural network (RNN)-based text encoder to mine the knowledge from RS images and texts, respectively. However, various studies show that CNNs pay more attention to local information and ignore the global clues hidden in images [14], [15], [16]. Also, the RNN-based text encoders face the difficulties of establishing long-range dependencies [17]. These shortcomings limit the CNN- and RNN-based encoders' performance in RSITR.

Recently, with the remarkable capacity in capturing long-range dependencies and global contextual information, Transformer [17] has become popular in natural language processing (NLP) [18], [19] and computer vision (CV) [20], [21]. Moreover, Transformer has also demonstrated great potential in the visual-linguistic (VL) task, especially for the image–text retrieval task in natural scene [22], [23]. This is attributed to its specific architecture, which can receive the combination of images and texts [24], [25], [26], [27]. This way, not only

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can the specific features corresponding to images and texts be captured but also the interactive cross-modal relations can be investigated.

Although the above image–text retrieval models have succeeded in their applications, it is challenging to generalize them to RS scenes directly. On the one hand, compared with the natural images, the contents within RS images are more complex [28], [29], [30]. Also, their scale clues are diverse [6], [31]. These characteristics enhance the difficulties of visual feature learning. Compared with simple textual descriptions, RS images contain abundant and diverse objects [32], [33]. These unbalanced semantic relationships significantly increase the difficulty of image–text matching (ITM), as all objects, apart from those described in the text, act as distractors for matching. Furthermore, the "false-negative (FN) samples" problem (i.e., different images have identical textual captions) always impacts image–text retrieval tasks negatively. Due to the high intraclass diversity and interclass similarity of RS images, the issue of FN samples is particularly prominent in RSITR, thus limiting the model's performance [8].

We propose a new Transformer-based RSITR method named prior-experience-based RS vision-language (PERSVL) model to overcome the mentioned obstacles. It encloses a text encoder, a multiscale vision encoder, a dual-branch multimodal fusion encoder (DBMFE), and a learning from prior experiences (LPEs) module. Two encoders aim to map texts and RS images into textual and visual features adequately. DBMFE is developed to fuse the features from two modalities by fully exploring their relationships. LPE focuses on alleviating the undesirable influence of FN samples by allocating reliable dynamic soft labels to them with the consideration of historical data. To ensure the functions of various components in PERSVL, several tasks are also designed, including image–text contrastive (ITC) learning, mask language modeling (MLM), mask image modeling (MIM), and ITM tasks. Together with the two-stage multitask training strategy, PERSVL can achieve satisfactory cross-modal retrieval results.

Our contributions can be summarized as follows.

1) An end-to-end RSITR model named PERSVL is proposed in this article. It is a Transformer-based network that is good at extracting global information and modeling relationships. The characteristics of PERSVL enable it to adapt to cross-modal learning naturally.
2) Considering that existing methods do not take into account the multiscale information and long-range relationships of RS images simultaneously, a high-level feature complement (HFC) module is developed and embedded into the visual encoder to enhance PERSVL's capacity for visual feature learning. By effectively integrating the multiscale features learned by the visual backbone, the complex contents of RS images can be explored fully.
3) Due to the inadequacies of existing methods in capturing the relationships between visual and textual features, we construct a dual-branch DBMFE to deeply capture the intricate relationships across RS images and texts. It consists of a dual-branch interaction (DBMI) module and a branch fusion module. By mutually interacting with the valuable clues from different modalities, the visual and textual features can be improved and ejected into a multimodal representation.
4) To mitigate the issues arising from FN samples, we introduce an LPE module to PERSVL. Unlike the common methods, which generate the soft pseudolabels by measuring the similarity between modalities or within the text space, LPE employs prior knowledge to transform the negative samples' hard labels into dynamic soft labels, further boosting the PERSVL's retrieval accuracy.

The rest of this article is organized as follows. Some published works are reviewed in Section II. The proposed PERSVL is explained in Section III. Section IV displays the experimental results and discussion. A short conclusion is exhibited in Section V.

II. Related Works

This section reviews the existing image–text retrieval methods for the readers. For clarity, we divided them into dual-encoder-based and fusion-encoder-based models according to their architecture.

A. Dual-Encoder-Based Methods

The models with dual-encoder architecture first employ two independent encoders to extract the image and text features, respectively. Then, some common similarity metrics (such as cosine distance) are applied to different features to achieve image–text retrieval. These methods are popular for their straightforward ideas and efficient retrieval process. Contrastive language-image pretraining model (CLIP) [22] is a representative method in this group. It uses ResNet [34] (or ViT [20]), and GPT-2 [35] as image and text encoders to encode multimodal features. Then, the image–text retrieval tasks are completed according to cosine similarity between cross-modal features. To enhance the representation of each modality and reduce the intermodality discrepancies, Li et al. [36] proposed MEMBER. This method selects the maximum value of each dimension in the image (text) representation as the self-embedding feature and stores it in a memory bank. In the training process, each modality uses the cross-attention mechanism to fuse information from the other modal in the memory bank. This way, MEMBER can achieve satisfactory features for cross-modal retrieval. Similar to MEMBER, collaborative two-stream vision-language pretraining model (COTS) [23] designs a token-level interaction strategy, in which a cross-modal mask modeling task is established to understand and capture semantic information in images and texts. Also, COTS adopts momentum contrastive learning [37] method to distinguish between relevant and irrelevant features, thereby improving the quality of feature representation. Unlike the MEMBER and COTS, vision and scene text aggregation framework (ViSTA) [38] focuses on improving the visual representation by introducing the scene text information. In this method, a shared fusion token is developed to exchange relevant information between visual images and scene texts for enhancing the visual representation.

The dual-encoder architecture is also widely used in RSITR. Deep bidirectional triplet network (DBTN) [7] leverages EfficientNet [39] and long short-term memory (LSTM) [40] to

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encode RS images and texts. Then, Euclidean distance is used to measure the similarities between different modal features. Combining the valuable features and credible resemblance, DBTN achieves positive RSITR results. Al Rahhal et al. [41] use a CLIP encoder with global average pooling as the backbone to accomplish RSITR. In addition, they propose a multilanguage training strategy to enhance the model's generalization. Considering the complex contents within RS images, GaLR [11] enhances the visual encoder to ensure the performance of RSITR. It first employs graph convolutional network (GCN) [42] to explore various objects and model their correlations. Then, the multilevel visual self-attention (MVSA) module [8] is used to investigate the multiscale information of RS images. Finally, the significant visual features can be obtained by integrating the intricate object-level correlations and multiscale clues. The encouraging RSITR results can be produced based on these visual features. Although the results of the above methods are promising, their computational costs are high. To mitigate this issue, Yuan et al. [43] proposed a lightweight model named lightweight RS crossmodal multiscale retrieval model (LW-MCR). It develops a bilinear fusion scheme and a knowledge distillation strategy to enable RSITR performance and reduce time costs simultaneously.

B. Fusion-Encoder-Based Methods

The models with the fusion-encoder architecture always design complex cross-modal interaction schemes to fully mine the relationships between images and texts. This way, the accuracy of image–text retrieval tasks can be ensured. Note that the approaches in this group achieve the matching scores of image and text pairs directly. For example, context-aware attention network (CAAN) [44] employs bottom–up attention [45] and Bi-gated recurrent unit (GRU) [46], [47] to extract image and text features, respectively, and uses inner product to calculate cross-modal similarity. To capture the correlations between images and texts, CAAN designs context-aware attention to assign fine-grained weights to different modalities. In this fashion, the image and text features can be aligned, which enhances the final retrieval results. Benefiting from the self-attention mechanism (SAM) of the Transformer, many cross-modal learning studies [24], [25], [26], [27] have been proposed to highlight the cross-modal interactions. As a typical one, vision-and-language transformer (ViLT) [24] first maps images and texts into embedding in an easy way. Then, it concatenates two kinds of embeddings and inputs them into the following Transformer encoders. This way, the intramodal information and intermodal relations can be fully explored for RSITR tasks. Like ViLT, another model was developed in [25] named align before fuse model (ALBEF). After getting the textual and visual embeddings, ALBEF first applies a cross-modal alignment strategy to whittle down the discrepancies between modalities effectively. Then, the aligned embeddings are fed into the following Transformer encoders. The useful representations and cross-modal relations can be mined along with the cross-attention and SAM mechanisms.

Due to the powerful ability to model cross-modal relations, the fusion-encoder architecture has become increasingly popular in RSITR. SAM [10] leverages conventional CNN and RNN to learn features from RS images and texts. Then, a semantic alignment module is introduced to investigate the cross-modal relations and enhance the features by a mutual supplement scheme. Therefore, the reliability of RSITR results can be ensured. In view of the complexity of RS images, apart from the cross-modal relations, researchers pay more attention to learning the visual features. For instance, Yuan et al. [8] presented asymmetric multimodal feature matching network (AMFMN) for RSITR. It designs a multiscale feature fusion module to explore complex contents within RS images. Also, a visual-guided attention module is developed to align cross-model semantics and improve the final retrieval. Based on AMFMN, cross-modal RS feature matching network (CMFM-Net) was proposed in [12]. With the help of GCN [42], the representations corresponding to different modalities are strengthened. The encouraging RSITR results can be achieved with the visual-guided attention module. To further model cross-modal relationships, interacting-enhancing feature transformer (IEFT) [48] uses Transformer encoders to jointly learn the RS images and texts' features and their relationships simultaneously. In addition, IEFT designs a channelwise feature-enhancing unit to enrich the visual representation for improving the retrieval lists.

III. Method

A. Overall Framework

The overall framework of PERSVL is illustrated in Fig. 1. It contains a text encoder, a vision encoder, a DBMFE module, and an LPE module. Suppose that there is an image–text pair $\{\mathbf{I}, \mathbf{T}\}$, where $\mathbf{I} \in \mathbb{R}^{C\times H\times W}$ represents an RS image and $\mathbf{T}$ (with arbitrary size) represents a text. When inputting them into our PERSVL, two encoders will be used to map them into different feature representations first. For $\mathbf{T}$, we employ bidirectional encoder representations from transformers (BERT) [18] as our text encoder to generate textual representation $\mathbf{F}^{\mathrm{Te}}$. For $\mathbf{I}$, a specific vision encoder is developed to explore the complex contents within the RS image, encompassing a Swin-Transformer [21] backbone and an HFC module. The Swin-Transformer is responsible for learning multiscale information from the RS image, and the HFC module is in charge of fusing those multiscale clues and producing the visual representation $\mathbf{F}^{I}$. Meanwhile, an ITC task is added to $\mathbf{F}^{I}$ and $\mathbf{F}^{\mathrm{Te}}$ for aligning the semantics corresponding to the different modalities. Then, the DBMFE module is applied to $\{\mathbf{F}^{I}, \mathbf{F}^{\mathrm{Te}}\}$ to produce the fusion feature $\mathbf{F}^{M}$. Two tasks, i.e., mask image and language modeling, are designed in DBMFE to accomplish the information complement between visual and textual modalities. Finally, $\mathbf{F}^{M}$ can be mapped into the matching score for RSITR tasks. Here, the LPE module is added to mitigate the adverse effects of FN samples by assigning dynamic soft labels to them. To train our PERSVL, we also present a multitask training strategy. By considering the mentioned tasks simultaneously, PERSVL's cross-modal interaction capability can be guaranteed.

B. Text and Vision Encoders

Compared with the popular RNN-based text encoders, Transformer-based text encoders [17], [18], [19] are good at modeling global relations hidden in the input data, which makes them more suitable for RSITR tasks. As a typical

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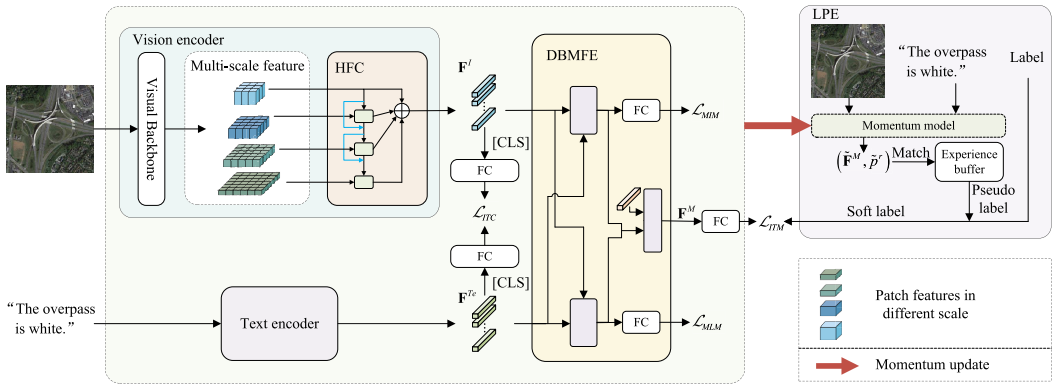

Fig. 1. Overall framework of PERSVL. It contains a vision encoder, a text encoder, a DBMFE, and an LPEs module. The vision encoder mainly encompasses a visual backbone (Swin Transformer) and an HFC module. It focuses on extracting the multiscale features from RS images and combining them effectively. Meanwhile, BERT is employed as our text encoder to generate textual representation. After obtaining the visual and textual features, DBMFE will be applied to enrich them and achieve a unified multimodal representation. Finally, a classification head composed of linear layers and a softmax function is used to generate ITM probabilities. Also, the LPE module is introduced in the training stage to mitigate the " FN samples" problem, thereby aiding the model in better learning the relationships between images and text.

example, BERT [18] has achieved significant success in NLP tasks due to its bidirectional sequence modeling manner and self-supervised pretraining design. Therefore, in this article, we employ BERT as the text encoder. To save the training time costs, followed by ALBEF [25], we use the first six layers of the BERT model as the text encoder to extract high-level semantic features. Specifically, given a text $\mathbf{T} = \{w_1, w_2, \ldots, w_k\}$ containing k words, the BERT tokenizer is first used to generate the word token sequence $\mathbf{T}' = \{\mathbf{t}_{\text{cls}}, \mathbf{t}_1, \ldots, \mathbf{t}_k, \mathbf{t}_{\text{sep}}\}$, where t_{cls} and t_{sep} represent the tokens corresponding to the special symbols "[CLS]" and "[SEP]," and $\mathbf{t}_i$ means the ith word's token. Then, passing $\mathbf{T}'$ through six Transformer layers, the textual representation $\mathbf{F}^{\text{Te}} = \{\mathbf{H}^{\text{cls}}, \mathbf{H}_1, \ldots, \mathbf{H}_k, \mathbf{H}^{\text{sep}}\} \in \mathbb{R}^{l \times d_l}$ can be obtained, where $\mathbf{H}^{\text{cls}}$ and $\mathbf{H}^{\text{sep}}$ denote the special symbols' representations, $\mathbf{H}_i$ implies the ith word's representation, $l = k + 2$ indicates the length of the textual feature, and d_l signifies the dimension of the textual representation.

Similarly, due to the capacity to explore long-distance dependencies, we select a Transformer-based visual encoder rather than the common CNN-based vision encoders as the visual backbone to extract the valuable clues from RS images. Considering the time complexity, we adopt the Swin Transformer [21] in this article. There are four stages in Swin-Transformer, each of which uses a patch merging module and a shifted window-based SAM to learn the global and local features from the input data. With the pyramid architecture and shifted window mechanism, Swin Transformer can significantly reduce computational costs while fully exploring multiscale features hidden in the input image. After passing an RS image $\mathbf{I} \in \mathbb{R}^{C \times H \times W}$ through Swin-Transformer, we can obtain the multiscale features $\mathbf{M}_1 \in \mathbb{R}^{(HW/4\times4)\times c}$, $\mathbf{M}_2 \in \mathbb{R}^{(HW/8\times8)\times2c}$, $\mathbf{M}_3 \in \mathbb{R}^{(HW/16\times16)\times4c}$, and $\mathbf{M}_4 \in \mathbb{R}^{(HW/32\times32)\times8c}$, where c represents the channel number of the output in the first stage.

The HFC module focuses on fusing the obtained multiscale features $\{\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3, \mathbf{M}_4\}$ to deeply capture the complex contents within RS images, whose architecture is shown in Fig. 2. To attain this objective, a multiscale attention mechanism (MSAM) is devised for hierarchically enhancing various

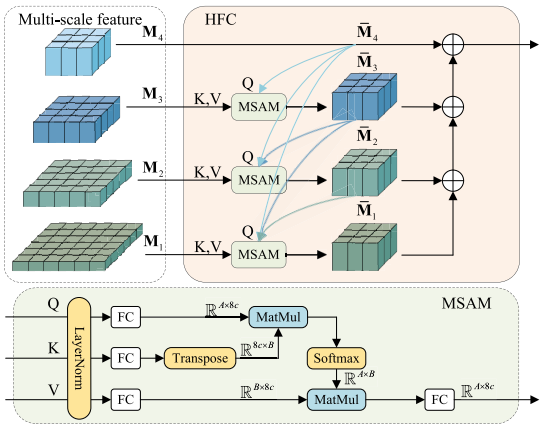

Fig. 2. Flowchart of HFC. It mainly incorporates an MSAM and a simple fusion scheme. MSAM hierarchically enhances various features by adding information from deeper layers. The simple fusion scheme is used to integrate the enhanced features.

features. For the level h (where $h = 1, 2,$ and 3), we set the complete feature in higher level $\sum_{l=h+1}^{4} \bar{\mathbf{M}}_{4-l}$ as a *Query*. Also, the hth level feature $\mathbf{M}_h$ is regarded as the *Key* and *Value*. Then, the enhanced $\bar{\mathbf{M}}_h$ can be calculated by

$$\begin{aligned}
\bar{\mathbf{M}}_h &= \left(\text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h{}^T}{\sqrt{8c}}\right)\mathbf{V}_h\right)\mathbf{W}_h^O \\
\mathbf{Q}_h &= \left(\text{LN}\left(\sum_{l=h+1}^{4} \bar{\mathbf{M}}_{4-l}\right)\right)\mathbf{W}_h^Q \\
\mathbf{K}_h &= (\text{LN}(\mathbf{M}_h))\mathbf{W}_h^K \\
\mathbf{V}_h &= (\text{LN}(\mathbf{M}_h))\mathbf{W}_h^V
\end{aligned} \tag{1}$$

where the matrices $\mathbf{W}_h^Q \in \mathbb{R}^{8c\times8c}$, $\mathbf{W}_h^K \in \mathbb{R}^{(8c/2^{4-h})\times8c}$, $\mathbf{W}_h^V \in \mathbb{R}^{(8c/2^{4-h})\times8c}$, and $\mathbf{W}_h^O \in \mathbb{R}^{8c\times8c}$ represent the linear projectors, softmax$(\cdot)$ indicates the nonlinear activation function, and LN$(\cdot)$ denotes the layer normalization function. This way, both the information at the current and deeper levels are thoroughly taken into account, thus guaranteeing the effectiveness and integrity of upgraded features. Note that we set $\bar{\mathbf{M}}_4 = \mathbf{M}_4$ in the above process. When the enhanced features $\{\bar{\mathbf{M}}_1, \bar{\mathbf{M}}_2, \bar{\mathbf{M}}_3, \bar{\mathbf{M}}_4\}$ are obtained, we aggregate them with a classification token $\mathbf{M}^{\text{cls}} \in \mathbb{R}^{8c}$ to generate the

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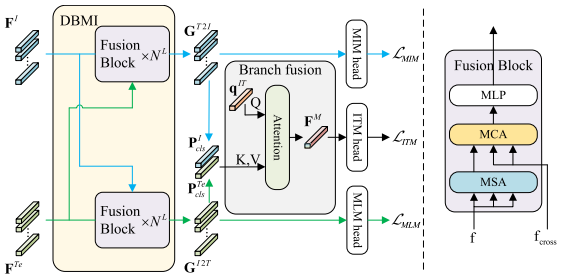

Fig. 3. Architecture of DBMFE, which consists of a DBMI module and a branch fusion module. DBMI involves the visual and textual branches, which aim to integrate visual (textual) information with textual (visual) clues. The branch fusion module fuses the knowledge from two modalities and generates a unified multimodal feature. The MLM, MIM, and ITM tasks are added to ensure the effectiveness of DBMFE.

comprehensive visual representation $\mathbf{F}^I = [\mathbf{M}^{\text{cls}}, \sum_{j=1}^{4} \bar{\mathbf{M}}_j] \in \mathbb{R}^{((HW/(32\times32))+1)\times 8c}$.

Although $\mathbf{F}^I$ and $\mathbf{F}^{\text{Te}}$ are discriminative in their modality, their misaligned semantics lead to the weak behavior in cross-modal understanding. To mitigate this problem, we introduce the ITC task [22], [25] for them. By pulling matched RS images and texts together and pushing the mismatched pairs away from each other, the semantics from different modalities can be aligned. Given N image–text feature pairs and their representations $\{\{\mathbf{F}_1^I, \mathbf{F}_1^{\text{Te}}\}, \ldots, \{\mathbf{F}_N^I, \mathbf{F}_N^{\text{Te}}\}\}$, the loss function the ITC task can be formulated as follows:

$$\begin{aligned}
\mathcal{L}_{\text{ITC}} &= \frac{1}{2}(\mathcal{L}_{I2T} + \mathcal{L}_{T2I}) \\
\mathcal{L}_{I2T} &= -\frac{1}{N}\sum_{i=1}^{N}\left[\log\frac{\exp\left(\mathbf{M}_i^{\text{cls}}\mathbf{H}_i^{\text{cls}^{\text{T}}}/\tau\right)}{\sum_{j=1}^{N}\exp\left(\mathbf{M}_i^{\text{cls}}\mathbf{H}_j^{\text{cls}^{\text{T}}}/\tau\right)}\right] \\
\mathcal{L}_{T2I} &= -\frac{1}{N}\sum_{i=1}^{N}\left[\log\frac{\exp\left(\mathbf{M}_i^{\text{cls}}\mathbf{H}_i^{\text{cls}^{\text{T}}}/\tau\right)}{\sum_{j=1}^{N}\exp\left(\mathbf{M}_j^{\text{cls}}\mathbf{H}_i^{\text{cls}^{\text{T}}}/\tau\right)}\right]
\end{aligned} \tag{2}$$

where τ is a learnable temperature parameter, $\mathbf{M}_i^{\text{cls}}$ and $\mathbf{H}_i^{\text{cls}}$ are the "[CLS]" token features of $\mathbf{F}_i^I$ and $\mathbf{F}_i^{\text{Te}}$, respectively. $\mathcal{L}_{I2T}$ is used to pull the RS images and their matched texts and push the mismatched texts away from them, while $\mathcal{L}_{T2I}$ is designed to pull the texts and their matched RS images and push the mismatched RS images away from them.

C. Dual-Branch Multimodal Fusion Encoder

The architecture of DBMFE is shown in Fig. 3. It comprises a DBMI module and a branch fusion module. DBMI enriches the visual/textual feature by interacting with the knowledge from the other modality, while the branch fusion module integrates the cross-modal features into a unified feature.

There are visual and textual branches in the DBMI module, each involving N^L fusion blocks. The structure of the fusion block follows the Transformer decoder [17], which encloses a multihead self-attention (MSA) layer, a multihead cross-attention (MCA), and a multilayer perception (MLP) layer. The MSA layer captures the correlation within the current modality, the MCA layer injects information from one modality to another, and the MLP layer integrates the clues obtained by MSA and MCA. Since all fusion blocks are the same, we take the ith block in the visual branch as an example to explain.

In detail, given the text representation $\mathbf{F}^{\text{Te}}$ and the output of $(i-1)$th block $\mathbf{G}_{i-1}^{T2I}$ in the visual branch, the output $\mathbf{G}_i^{T2I}$ of ith block is calculated as follows:

$$\begin{aligned}
\mathbf{G}_i^{T2I} &= \text{MLP}\left(\bar{\mathbf{G}}_i^{T2I}\right) + \bar{\mathbf{G}}_i^{T2I} \\
\bar{\mathbf{G}}_i^{T2I} &= \text{MCA}\left(\bar{\mathbf{M}}_i^{T2I}, \mathbf{F}^{\text{Te}}, \mathbf{F}^{\text{Te}}\right) + \bar{\mathbf{M}}_i^{T2I} \\
\bar{\mathbf{M}}_i^{T2I} &= \text{MHA}\left(\mathbf{G}_{i-1}^{T2I}, \mathbf{G}_{i-1}^{T2I}, \mathbf{G}_{i-1}^{T2I}\right) + \mathbf{G}_{i-1}^{T2I}
\end{aligned} \tag{3}$$

where $\bar{\mathbf{M}}_i^{T2I}$ and $\bar{\mathbf{G}}_i^{T2I}$ are the two intermediate feature maps with the size of $((HW/(32\times32)))\times d$,[1] and MSA$(\cdot)$, MCA$(\cdot)$, and MLP$(\cdot)$ are the functions of MSA, MCA, and MLP layers, whose definitions are followed by the standard Transformer [17]. After N^L fusion blocks, we can obtain the visual mixture features $\mathbf{G}^{T2I} \in \mathbb{R}^{((HW/(32\times32))+1)\times d}$. As the same operation, the textual mixture features $\mathbf{G}^{I2T} \in \mathbb{R}^{L\times d}$ can also be produced.

To further enhance the cross-modal interaction capacity of DBMI, we design an MLM task and an MIM task. They mask partial information of one modality to force the DBMI module to deeply explore the valuable clues from another modality. Particularly, we first construct the masked input RS image and text pair $\{\mathbf{I}^m, \mathbf{T}^m\}$ by randomly masking the pixels and the nouns, adjectives, and quantifiers. In this article, the visual and textual masked rates are 60% and 15%, resectively. Also, we suppose their features $\{\mathbf{F}^{Im}, \mathbf{F}^{Tm}\}$ have been obtained by the visual and textual encoders. For the visual branch in DBMI, the masked visual mixture feature $\mathbf{G}^{T2Im}$ can be generated by passing $\mathbf{F}^{Im}$ and $\mathbf{F}^{\text{Te}}$ through N^L fusion blocks. Similarly, the masked textural mixture feature $\mathbf{G}^{I2Tm}$ can also be produced in the DBMI's textual branch. Then, an MIM head is introduced to map $\mathbf{G}^{T2Im}$ into the prediction $\mathbf{x}^{im}$ of masked pixels. Likewise, an MLM head is also applied on $\mathbf{G}^{I2Tm}$ to create the probability distribution $\mathbf{p}^{tm} \in \mathbb{R}^{N^m\times D^t}$, where N^m is the number of masked tokens, D^t is the size of text vocabulary. Two heads enclose fully connected (FC) and softmax activation layers.

Based on the above explanations, the loss functions of MLM and MIM tasks can be defined as follows. Suppose there are N pairs of masked RS images and texts, and their masked pixels' predictions and tokens' probability distributions are $\{\mathbf{x}_1^{im}, \ldots, \mathbf{x}_N^{im}\}$ and $\{\mathbf{p}_1^{tm}, \ldots, \mathbf{p}_N^{tm}\}$. The MLM and MIM loss functions are

$$\begin{aligned}
\mathcal{L}_{\text{MIM}} &= -\frac{1}{N}\sum_{j=1}^{N}\frac{\left\|\mathbf{y}_j^{im} - \mathbf{x}_j^{im}\right\|_1}{\Omega\left(\mathbf{y}_j^{im}\right)} \\
\mathcal{L}_{\text{MLM}} &= -\frac{1}{N}\sum_{j=1}^{N}\frac{\left(\log\mathbf{p}_j^{tm}\right)\times\left(\mathbf{y}_t^{tm}\right)^T}{N^m}
\end{aligned} \tag{4}$$

where $\mathbf{y}_j^{im}$ and $\Omega(\mathbf{y}_j^{im})$ represent the values and volume of masked pixels in the jth masked RS image, and $\mathbf{y}_j^{tm} \in \mathbb{R}^{D^t}$ denotes the one-hot label for the masked words in the jth masked text.

The other part of DBMFE is the branch fusion module. It leverages a weighted strategy to fuse the mixture features $\mathbf{G}^{T2I}$ and $\mathbf{G}^{I2T}$ to generate a unified multimodal representation

[1]Henceforth, we represent $8c$ and d_l as d since they are equivalent, i.e., $8c = d_l$.

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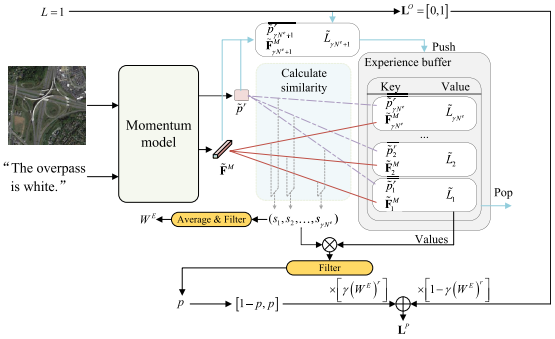

Fig. 4. Flowchart of the LPE module. It leverages the momentum PERSVL to generate the prior experience and transform the hard labels of negative samples into soft labels. This way, the FN samples problem can be mitigated.

$\mathbf{F}^M \in \mathbb{R}^d$. Specifically, given the "[CLS]" token features $\mathbf{P}_{\text{cls}}^V \in \mathbb{R}^d$ and $\mathbf{P}_{\text{cls}}^{\text{Te}} \in \mathbb{R}^d$ of $\mathbf{G}^{T2I}$ and $\mathbf{G}^{I2T}$, we first concatenate them as the joint multimodal vectors $\mathbf{F}^{IT} \in \mathbb{R}^{2\times d}$. Then, the attention mechanism is applied to $\mathbf{F}^{IT}$ with a learnable *Query* $\mathbf{q}^{IT} \in \mathbb{R}^{1\times d}$ for the fusion. This step can be formulated as follows:

$$\begin{aligned}\mathbf{F}^M &= \left(\text{softmax}\left(\frac{\mathbf{Q}^{IT}\mathbf{K}^{IT}}{\sqrt{d}}\right)\mathbf{V}^{IT}\right)\mathbf{W}_m^O + \mathbf{q}^{IT} \\ \mathbf{Q}^{IT} &= \mathbf{q}^{IT}\mathbf{W}_Q^{IT}, \mathbf{K}^{IT} = \mathbf{F}^{IT}\mathbf{W}_K^{IT}, \mathbf{V}^{IT} = \mathbf{F}^{IT}\mathbf{W}_V^{IT}\end{aligned} \tag{5}$$

where $\mathbf{W}_Q^{IT}$, $\mathbf{W}_K^{IT}$, $\mathbf{W}_V^{IT}$, and $\mathbf{W}_m^O$ are the learnable matrix with a size of $d \times d$. In (5), $\mathbf{q}^{IT}$ serves as the fundamental component for generating dynamic fusing weights, enabling the branch fusion module to flexibly allocate importance to $\mathbf{G}^{T2I}$ and $\mathbf{G}^{I2T}$. Based on the adoption of $\mathbf{q}^{IT}$, our branch fusion module can preserve the inherent distinctions within the individual features while simultaneously capturing and characterizing the intricate relationships between two modalities.

When $\mathbf{F}^M$ is gotten, we can use it to predict the matching score between RS image and text for the final retrieval. To this end, an ITM head that contains an FC layer and a softmax activation is applied on $\mathbf{F}^M$ to generate a 2-D image–text probability distribution $\mathbf{p}^{\text{itm}}$, where $\mathbf{p}^{\text{itm}}(0)/\mathbf{p}^{\text{itm}}(1)$ indicates the mismatching/matching score between RS image and text. To enhance the reliability of $\mathbf{p}^{\text{itm}}$, an ITM task is designed. Suppose that there are N image–text pairs, and their image–text probability distributions and labels are $\{\mathbf{p}_1^{\text{itm}}, \ldots, \mathbf{p}_N^{\text{itm}}\}$ and $\{L_1, \ldots, L_N\}$, the loss function of the ITM task is

$$\mathcal{L}_{\text{ITM}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\left(\log \mathbf{p}_i^{\text{itm}}\right) \times \mathbf{L}_i{}^{\text{T}}\right] \tag{6}$$

where $\mathbf{L}_i = [1 - L_i, L_i]$ denotes the 2-D version of ith label L_i. Note that the value of L_i is either 0 or 1. When $L_i = 0$, it indicates that the RS image and text in the ith pair do not match, and vice versa.

D. Learning From Prior Experiences

To conquer the FN sample problem, we develop an LPE module and add it to our PERSVL in the training process, whose flowchart is exhibited in Fig. 4. It transforms hard labels of the negative samples into reliable soft labels based on prior experiences to mitigate the influence of FN samples. To achieve this goal, LPE adopts a momentum theory [37] to generate and store the experiences. The structure of the momentum model is the same as our PERSVL. However, the parameters of momentum PERSVL are inherited from PERSVL, i.e.,

$$\theta_m = \eta\theta_m + (1 - \eta)\theta \tag{7}$$

where θ_m and θ represent the parameters of the momentum PERSVL and the PERSVL model, $\eta \in (0, 1)$ is a hyperparameter. After feeding RS image and text pairs into the momentum PERSVL sequentially, a series of multimodal fusion features and matching scores can be generated. We call them experience and store them into a buffer $\mathcal{B}$. The definition of jth experience item is

$$\mathbf{E}_j = \left\{\tilde{\mathbf{F}}_j^M, \tilde{p}_j^r, \tilde{L}_j\right\} \tag{8}$$

where $\tilde{\mathbf{F}}_j^M$ indicates the multimodal fusion feature, $\tilde{p}_j^r$ implies the matching score [i.e., $\tilde{\mathbf{p}}_j^{\text{itm}}(1)$], and $\tilde{L}_j$ denotes the hard label. The buffer $\mathcal{B}$ is a queue with a set of experiences. When it is full, the oldest experience would be removed. Once an experience is added into $\mathcal{B}$, it will be regarded as a prior experience. Another point we want to touch on is that the hard label $\tilde{L}_j$ is adjusted slightly in LPE. In particular, the hard label of the RS image and its least similar text in a mini-batch is set to -1 instead of 0. This way, the confidence of prior experience in $\mathcal{B}$ will be enhanced.

Now, assuming that there are N^e prior experience items in $\mathcal{B}$, i.e., $\mathcal{B} = \{\mathbf{E}_1, \ldots, \mathbf{E}_{N^e}\}$. For a current experience $\mathbf{E}$, if its hard label $L = 1$, we store it into $\mathcal{B}$ directly. Otherwise, the hard label will be transformed into the soft label. Specifically, we first design a hybrid similarity measure to decide the resemblance between it and the prior experience $\mathbf{s} = \{s_1, \ldots, s_{N_e}\}$. The definition of s_j is

$$s_j = \left(\tilde{p}^r - \tilde{p}_j^r\right)^2 \text{Filter}\left(\cos\left(\tilde{\mathbf{F}}^M, \tilde{\mathbf{F}}_j^M\right), \alpha\right) \tag{9}$$

where $\cos(\cdot)$ means the cosine similarity, α is the threshold, and $\text{Filter}(\cdot, \alpha)$ denotes a filter function whose definition is

$$\text{Filter}(x, \alpha) = \begin{cases} x, & x \geq \alpha \\ -1, & x < \alpha. \end{cases} \tag{10}$$

Then, the pseudolabel p of $\{\mathbf{I}, \mathbf{T}\}$ is generated by

$$p = \left(\sum_{j=1}^{N^e} s_j \tilde{L}_j\right)\mathbb{I}\left(\sum_{j=1}^{N^e} s_j \tilde{L}_j > 0\right) \tag{11}$$

where $\mathbb{I}(\cdot)$ is indicator function. After that, the pseudolabel's 2-D version can be formulated as $\mathbf{L}^E = [1 - p, p]$. Next, the soft label of $\{\mathbf{I}, \mathbf{T}\}$ can be calculated by

$$\begin{aligned}\mathbf{L}^S &= c\mathbf{L}^E + (1 - c)\mathbf{L}^O, c = \gamma \times \left(W^E\right)^r \\ W^E &= \frac{\left(\mu^S - \alpha\right)}{1 - \alpha}\mathbb{I}\left(\mu^S > \alpha\right) \\ \mu^S &= \frac{1}{N^e}\sum_{j}^{N^e}\text{C}\left(\tilde{\mathbf{F}}^M, \tilde{\mathbf{F}}_j^M\right)\end{aligned} \tag{12}$$

where r is an exponentially learnable parameter, γ is the ratio of the number of prior experience in $\mathcal{B}$ to the capacity of $\mathcal{B}$,

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and $\mathbf{L}^O = [1-L, L]$ is the original label. When we get $\mathbf{L}^S$, the ITM loss function [i.e., (6)] can be rewritten as follows:

$$\mathcal{L}_{\mathrm{ITM}}^{S} = -\frac{1}{N}\sum_{i=1}^{N}\left[\left(\log \mathbf{p}_i^{\mathrm{itm}}\right) \times \mathbf{L}_i^{S^T}\right]. \tag{13}$$

In addition, to better balance $\mathbf{L}^E$ and $\mathbf{L}^O$, we also propose a balance parameter restriction (BPR) loss, which minimizes the gap between the ITM loss calculated using $\mathbf{L}^E$ and $\mathbf{L}^O$, i.e.,

$$\begin{aligned}\mathcal{L}_{\mathrm{BPR}} &= \mathrm{MSE}\left(\mathcal{L}_{\mathrm{ITM}}^{O}(\mathbf{I},\mathbf{T}), \mathcal{L}_{\mathrm{ITM}}^{P}(\mathbf{I},\mathbf{T})\right)\\ \mathcal{L}_{\mathrm{ITM}}^{O}(\mathbf{I},\mathbf{T}) &= -\left(\log \mathbf{p}'^{\mathrm{itm}}\right) \times \left(\mathbf{L}^O\right)^T\\ \mathcal{L}_{\mathrm{ITM}}^{P}(\mathbf{I},\mathbf{T}) &= -\left(\log \mathbf{p}'^{\mathrm{itm}}\right) \times \left(\mathbf{L}^S\right)^T\end{aligned} \tag{14}$$

where $\mathbf{p}'^{\mathrm{itm}}$ represents the probability distribution without gradient and MSE$(\cdot)$ represent the mean squared error loss function.

E. Two-Stage Training Strategy

We propose a two-stage multitask training strategy to fully leverage the contributions of different tasks to our PERSVL. In the first stage, the ITM, ITC, MIM, and MLM tasks are used to train our PERSVL with the original hard labels. In this fashion, the preliminary cross-modal matching capacity of PERSVL can be achieved. In the second stage, besides the above four tasks, the LPE module will be added to enhance the model's performance, and the BPR task will be added simultaneously to assist the LPE module in generating more robust soft pseudolabels. The overall loss functions of the two stages are

$$\begin{aligned}\mathcal{L}^{s1} &= \mathcal{L}_{\mathrm{ITM}} + \lambda_1\mathcal{L}_{\mathrm{ITC}} + \lambda_2(\mathcal{L}_{\mathrm{MIM}} + \mathcal{L}_{\mathrm{MLM}})\\ \mathcal{L}^{s2} &= \mathcal{L}_{\mathrm{ITM}}^{S} + \lambda_1\mathcal{L}_{\mathrm{ITC}} + \lambda_2(\mathcal{L}_{\mathrm{MIM}} + \mathcal{L}_{\mathrm{MLM}}) + \lambda_3\mathcal{L}_{\mathrm{BPR}}\end{aligned} \tag{15}$$

where λ_1–λ_3 are the hyperparameters to control the contributions of different types of loss terms. With the help of multiple tasks and the two-stage training strategy, the proposed PERSVL can be fully trained.

IV. Experiment

A. Datasets and Evaluation Metrics

We use the following three datasets to assess our PERSVL.

1) *RS image captioning dataset (RSICD) [49]:* This is a large-scale RS caption dataset containing 10 921 RS images with a fixed size of 224 × 224. Their resolutions are various. There are 24 333 sentences with 3323 words in this dataset. Each RS image has five caption sentences. However, only 724 RS images have five distinct captions, while the others have repetitive sentences. The high caption repetition and diverse RS image resolutions increase the difficulty of this dataset.
2) *RSITMD Dataset [8]:* This dataset contains 4743 RS images with a size of 256 × 256. The RS images are sourced from the RSICD dataset and Google Earth. The number of captions is 23 715. Compared with the RSICD dataset, the sentences in this dataset contain more fine-grained information, as it pays more attention to the fine-grained relationship between objects in the annotation process. Similarly, each RS image has five captions.
3) *UCM-Captions Dataset [50]:* This dataset is built upon the UCMerced scene dataset [51], which contains 2100 RS images with a size of 256 × 256 and a resolution of 1 foot. The researchers provide five captions for each RS image. Thus, there are 10 500 sentences in this dataset. However, the repetition rate of captions is high, which significantly challenges the RSITR models.

In the following experiments, we regard RS images and their captions as positive samples. The rest of the combinations are considered as negative pairs.

To test the retrieval performance of the model, we used the following two metrics: $R@k$ and mR [48]. $R@k$ denotes the ratio of ground truth included in the top-k retrieval list. We use $R@1$, $R@5$, and $R@10$ as the evaluation metrics for image and text retrieval tasks. In addition, the metric mR (i.e., the mean value of the above six metrics) is used as an overall performance measure for the model. Note that, considering the duplicated captions, the image–text pairing relationships are simplified when we count the results to make the results more objective. Specifically, following by Tang et al. [48], we only consider whether a sentence is a caption of an RS image while ignoring which RS image it belongs to. This rule applies to PERSVL and all compared models.

B. Implementation Details

All experiments are conducted on NVIDIA RTXA6000 GPUs. For each dataset, we use 90% of the data to train PERSVL, and the remaining 10% data are adopted as the test set. The visual and textual encoders are initialized by the pretrained parameters, and other parts of our model are randomly initialized. In the training phase, the AdamW optimizer with an initial learning rate of 2×10^{-5} and weight decay of 2×10^{-2} is chosen to optimize the parameters. The training epoch is set to be 135 empirically. For RSITMD, two training stages are carried out in 70 and 65 epochs, respectively. For the other two datasets, the epochs for the two training stages are 50 and 85.

For texts, we unify $L = 30$ and $d_l = 768$ following the settings in BERT-Base [18]. The padding operation will be used if a textual feature's length is less than 30. For RS images, we resize them into 224 × 224 according to the structure of Swin-S [21]. In addition, the channel number of the output in the first stage of Swin-S equals 96, i.e., $c = 96$. Several parameters should be set in advance, including the number of blocks N_l in the DBMI module, the hyperparameter η in (7), the volume of buffer N^e, the similarity threshold α in (9) and (10), the exponentially learnable parameter r in (12), and the free parameters $\{\lambda_1, \lambda_2, \lambda_3\}$ in (15). Acoording to [25], we set $N^e = 65536$ and $\eta = 0.995$. Since r can be adjusted during training, we initialize it as 0.05. For others, the empirically optimal values are $N_l = 3$, $\alpha = 0.86$, $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = 1$. Section IV-E will discuss their influence on PERSVL.

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TABLE I
PERFORMANCE OF DIFFERENT MODELS COUNTED ON THE RSICD DATASET (%)

Model	Text Retrieval			Image Retrieval			mR
	$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$	
AMFMN-soft [8]	8.13	20.94	30.83	6.82	23.30	35.72	20.96
AMFMN-fusion [8]	9.39	20.61	31.34	6.61	22.77	35.22	20.99
AMFMN-sim [8]	8.39	21.22	30.49	7.51	23.49	36.72	21.30
GaLR [11]	8.35	20.49	30.49	6.43	23.64	37.18	21.10
MLT [41]	4.76	15.92	23.79	4.26	17.51	31.64	16.31
CMFM-Net [12]	9.15	21.77	32.02	7.92	24.17	37.68	22.12
IEFT [48]	8.95	24.50	38.29	5.08	21.64	37.83	22.72
KAMCL [52]	10.30	25.07	38.33	9.59	30.36	45.20	26.47
ViLT [24]	5.95	22.60	35.86	5.19	22.82	37.14	21.59
ALBEF [25]	8.51	22.41	33.94	7.14	25.21	40.81	23.00
BLIP [26]	7.86	21.68	32.57	6.18	22.76	37.26	21.39
PERSVL (Ours)	**10.70**	**27.08**	**38.43**	**10.14**	**30.43**	**45.58**	**27.06**

TABLE II
PERFORMANCE OF DIFFERENT MODELS COUNTED ON THE RSITMD DATASET (%)

Model	Text Retrieval			Image Retrieval			mR
	$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$	
AMFMN-soft [8]	12.53	31.11	44.09	10.70	36.56	55.45	31.74
AMFMN-fusion [8]	13.12	30.82	42.91	10.82	36.17	53.24	31.18
AMFMN-sim [8]	12.68	30.52	43.03	10.05	36.57	56.16	31.50
GaLR [11]	13.12	30.45	42.92	10.95	35.19	53.21	30.97
MLT [41]	8.40	18.14	31.19	7.43	29.38	51.33	24.31
CMFM-Net [12]	12.38	31.85	44.46	12.03	37.52	55.08	32.22
IEFT [48]	11.50	31.65	44.26	9.29	33.20	52.81	30.45
KAMCL [52]	15.04	31.93	45.21	13.70	40.65	57.95	34.08
ViLT [24]	9.96	31.19	44.47	9.10	32.61	52.50	29.97
ALBEF [25]	12.61	33.62	46.46	11.02	41.68	58.10	33.92
BLIP [26]	14.82	32.74	46.46	12.30	39.12	55.44	33.48
PERSVL (Ours)	**21.68**	**42.48**	**57.30**	**15.44**	**47.39**	**65.66**	**41.66**

C. Comparison With Existing Methods

We select the following 11 methods to verify the effectiveness of our model, they are, AMFMN-soft [8], AMFMN-fusion [8], AMFMN-sim [8], GaLR [11], multilanguage Transformer (MLT) [41], CMFM-Net [12], IEFT [48], knowledge-aided momentum contrastive learning method (KAMCL) [52], ViLT [24], ALBEF [25], and bootstrapping language image pretraining for unified vision-language understanding and generation (BLIP) [26]. The first eight methods are RS-oriented, and the last three are general-purpose. For fairness, the compared methods do not load their multimodal pretraining weights in the following experiments.

The performance of all methods counted on RSICD is summarized in Table I. By analyzing the results, we can easily find that the proposed PERSVL outperforms other methods in all cases. The enhancements achieved by PERSVL in mR are 6.10% (over AMFMN-soft), 6.07% (over AMFMN-fusion), 5.76% (over AMFMN-sim), 5.96% (over GaLR), 10.75% (over MLT), 4.94% (over CMFM-Net), 4.34% (over IEFT), 0.59% (over KAMCL), 5.47% (over ViLT), 4.06% (over ALBEF), and 5.67% (over BLIP). The reasons behind achieving such good results can be summarized in three points. First, the vision encoder in PERSVL can effectively explore relationships among the features from different levels, facilitating an efficient fusion of multiscale features. Thus, a more comprehensive visual representation can be provided for the downsteaming operations. Second, DBMFE fuses the cross-modal features into a unified feature and enriches the visual and textual features by deeply interacting with the useful clues hidden in them. This way, a more robust multimodal representation can be generated for RSITR tasks. Third, the problem of FN samples can be mitigated by our LPE module, which transforms the hard labels of negative samples into soft labels by considering prior experiences. Therefore, the performance of PERSVL is further enhanced.

Table II shows the results of all methods counted on RSITMD. Similar to the conclusions obtained from RSICD, the developed PERSVL is the best model among all methods. Taking "text retrieval" as an example, the improvements achieved by PERSVL in $R@1$ are 9.15% (over AMFMN-soft), 8.56% (over AMFMN-fusion), 9.00% (over AMFMN-sim), 8.56% (over GaLR), 13.28% (over MLT), 9.30% (over CMFM-Net), 10.18% (over IEFT), 6.64% (over KAMCL), 11.72% (over ViLT), 9.07% (over ALBEF), and 6.86% (over BLIP). Taking "image retrieval" as another instance, the enhancements in $R@1$ are 4.74% (over AMFMN-soft), 4.62% (over AMFMN-fusion), 5.39% (over AMFMN-sim), 4.49% (over GaLR), 8.01% (over MLT), 3.41% (over CMFM-Net), 6.15% (over IEFT), 1.74% (over KAMCL), 6.34% (over ViLT), 4.42% (over ALBEF), and 3.14% (over BLIP). In addition, the performance gaps between our model and the compared methods are distinct. Among all comparison algorithms, BLIP performed the best. However, the mR score of PERSVL is still 8% points higher than it. These positive results are mainly attributed to our model's thorough exploration of RS images and in-depth analysis of different modal data.

TABLE III
PERFORMANCE OF DIFFERENT MODELS COUNTED ON THE UCM-CAPTIONS DATASET (%)

Model	Text Retrieval			Image Retrieval			mR
	$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$	
AMFMN-soft [8]	43.81	61.42	72.85	54.47	78.28	88.09	66.49
AMFMN-fusion [8]	46.47	64.76	75.80	51.12	80.60	89.61	68.06
AMFMN-sim [8]	41.90	63.33	69.52	51.04	75.23	84.47	64.25
GaLR [11]	41.42	61.42	71.90	49.42	78.66	86.28	64.85
MLT [41]	40.95	63.81	77.62	53.14	82.29	96.10	68.98
CMFM-Net [12]	42.38	60.00	75.23	53.04	78.66	92.19	66.92
IEFT [48]	45.76	67.69	76.05	55.44	82.17	90.83	69.66
KAMCL [52]	43.17	55.87	67.77	51.17	80.15	88.69	64.47
ViLT [24]	45.71	61.43	75.24	54.23	83.78	91.94	68.72
ALBEF [25]	44.76	61.95	75.66	52.85	77.71	87.52	66.74
BLIP [26]	42.90	67.71	76.19	55.67	83.00	91.38	69.48
PERSVL (Ours)	**46.66**	**68.57**	**76.19**	**56.71**	**83.28**	**91.61**	**70.50**

The behavior of all methods counted on UCM-Captions is displayed in Table III. Unsurprisingly, PERSVL still achieves the best performance. The improvements obtained by PERSVL in mR are 4.01% (over AMFMN-soft), 2.44% (over AMFMN-fusion), 6.25% (over AMFMN-sim), 5.65% (over GaLR), 1.52% (over MLT), 3.58% (over CMFM-Net), 0.84% (over IEFT), 6.03% (over KAMCL), 1.78% (over ViLT), 3.76% (over ALBEF), and 1.02% (over BLIP). Nevertheless, an interesting observation is that the enhancements obtained by our model are not as high as the other two datasets. This is because UCM-Captions involves a limited volume of RS images. Also, the similarity in textual captions is relatively high. Overcoming the above two challenges and further improving PERSVL's performance are our future works.

Apart from the above scores, we exhibit the parameter volume of different methods in Table IV. We can find that PERSVL is lighter than ALBEF and BLIP but heavier than other models. After analysis, the two largest components in PERSVL regarding parameter volume are HFC and DBMFE.

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TABLE IV
PARAMETER VOLUME OF DIFFERENT MODELS

Model	AMFMN [8]	GaLR [11]	MLT [41]	CMFM-Net [12]	IEFT [48]	KAMCL [52]	ViLT [24]	ALBEF [25]	BLIP [26]	PERSVL (Ours)
Params (M)	36.70	46.90	149.62	51.15	100.12	124.18	111.60	209.60	223.70	184.93

TABLE V
ABLATION STUDIES ON EACH COMPONENT OF PERSVL (%). THE RESULTS ARE COUNTED ON RSITMD

Baseline	DBMFE	HFC	LPE	Text Retrieval			Image Retrieval			mR
				$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$	
√				19.69	39.78	56.08	14.03	46.09	64.64	40.05
√	√			19.91	40.24	56.63	15.23	46.15	64.86	40.50
√	√	√		21.37	42.28	57.02	15.25	47.07	65.61	41.43
√	√	√	√	**21.68**	**42.48**	**57.30**	**15.44**	**47.39**	**65.66**	**41.66**

HFC helps the visual encoder capture complex contents within RS images, and DBMFE utilizes a dual-branch structure to fuse cross-modal features fully. With their combined efforts, our model can achieve such encouraging performance. Thus, compared with the performance improvement, the parameter increase is worthwhile.

D. Ablation Studies

In this section, we conduct ablation experiments to validate each component's contributions to PERSVL. For clarity, we only use the RSITMD dataset.

1) Ablation Studies on the Whole Model: As discussed in Section III, the main components of the proposed PERSVL are HFC in the vision encoder, DBMFE, and LPE. To investigate their functions, we first construct a "Baseline" using the text encoder (i.e., BERT [18]) and Swin-Transformer [21]. When the different features are learned from texts and RS images, the ITC loss [22], [25] is adopted to train the "Baseline" and generate the retrieval results. Then, we add DBMFE, HFC, and LPE to "Baseline" to study their effects. The results are exhibited in Table V. The increasing trend in various assessment criteria indicates that the model's overall performance improves as the modules are stacked. Specifically, for mR, when adding the DBMFE to the "Baseline," we obtain an improvement with $+0.45\%$. This indicates that DBMFE effectively models the relationship between RS images and texts, thereby enhancing the model's performance. When adding the HFC module further, mR is enhanced with $+1.38\%$ compared with the "Baseline." This suggests that HFC adequately integrates multiscale information from RS images. The model's performance is further improved by obtaining a more comprehensive representation of RS images. When adding all the modules (i.e., DBMFE, HFC, and LPE) to the "Baseline," the performance is improved by $+1.61\%$. This demonstrates that the pseudolabels generated by the LPE module effectively address the problem of FN samples, thus aiding the model in better learning the relationships between images and text. These results demonstrate that each component plays a positive role in PERSVL.

2) Ablation Studies on the DBMFE Module: In this section, we investigate the function of DBMI and the branch fusion module in DBMFE. To this end, we construct two DBMFE variants. First, a "single" variant is constructed to illustrate the importance of DBMI. Consistent with [25], [26], the "single" variant retains the textual branch. In this case, the branch fusion module is replaced by a self-attention layer. The second variant combines DBMI and the "Avg Fusion" scheme to demonstrate the value of the branch fusion module. Here, "Avg Fusion" uses the averaging operation to combine different features. The performance of PERSVLs with these two variants is exhibited in Table VI. Also, the results of PERSVL with original DBMFE are exhibited for reference. The results are shown in Table VI. Compared with "single", our DBMFE achieves significant improvement for mR with $+1.72\%$. This reveals the superiority of DBMI. For the fusion strategies, the DBMFE with branch fusion module outperforms "Avg fusion" in all cases. This suggests that compared with the average fusion strategy, using a task-specific learnable query to generate dynamic weights is more conducive to merging the features generated by the DBMI module.

TABLE VI
ABLATION EXPERIMENTS ON DBMFE (%). THE RESULTS ARE COUNTED ON RSITMD DATASET

Model	Text Retrieval			Image Retrieval			mR
	$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$	
PERSVL+Single	18.14	40.48	54.64	14.95	46.41	65.04	39.94
PERSVL+DBMI+Avg fusion	19.91	42.47	56.19	15.04	45.88	65.35	40.81
PERSVL+DBMFE	**21.68**	**42.48**	**57.30**	**15.44**	**47.39**	**65.66**	**41.66**

TABLE VII
ABLATION EXPERIMENT ON LPE (%). THE RESULTS ARE COUNTED ON RSITMD DATASET

Model	Text Retrieval			Image Retrieval			mR
	$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$	
PERSVL+LPE w. cos	21.56	42.36	57.03	15.31	47.14	65.57	41.50
PERSVL+LPE w.o. BPR task	21.23	42.25	55.42	14.69	47.21	65.54	41.06
PERSVL+LPE	**21.68**	**42.48**	**57.30**	**15.44**	**47.39**	**65.66**	**41.66**

3) Ablation Studies on the LPE Module: As introduced in Section III-D, there are two key points in LPE. One is the hybrid similarity measure [see (9)], and the other is the BPR loss [see (14)]. Therefore, to deeply study LPE, we construct its two variants. The first variant is "LPE w. cos," in which the hybrid similarity measure is replaced with a cosine similarity. The second is "LPE w.o. BPR," where the BPR loss is removed from LPE. The performance of PERSVLs with them is displayed in Table VII. The results of PERSVL with the original LPE are also exhibited for reference. Compared with "LPE w. cos," our LPE achieves an improvement for mR with $+0.16\%$. This demonstrates that using hybrid similarity

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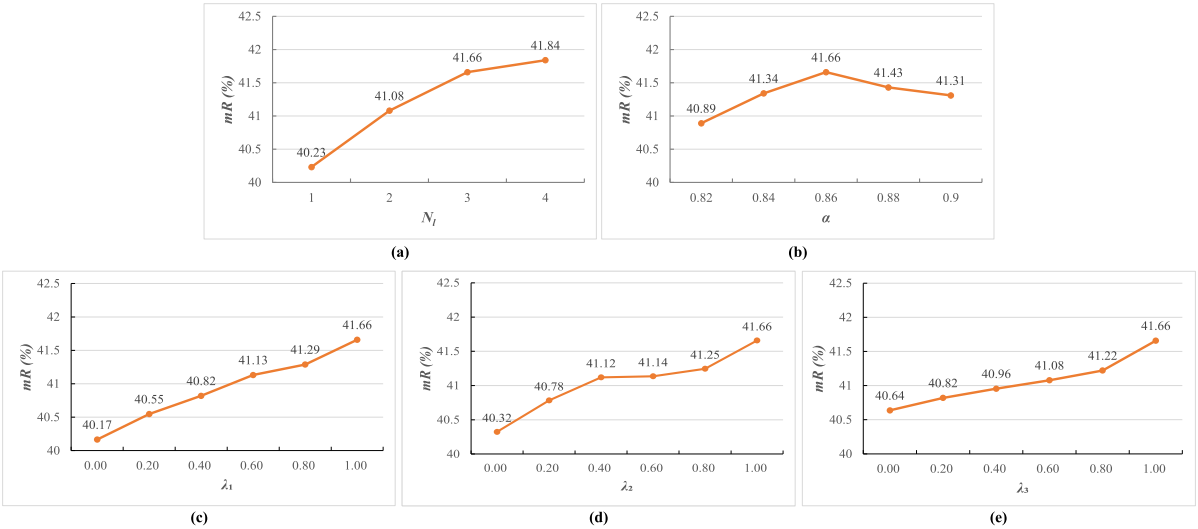

Fig. 5. Influence of free parameters on PERSVL. The mR scores are counted on RSITMD. (a) N_l, (b) α, (c) λ_1, (d) λ_2, and (e) λ_3.

can further measure the relationship between experiences. For the BPR loss, we observe that the LPE module with BPR loss outperforms "LPE w.o. BPR" with +0.60% for mR. This reveals BPR loss's effectiveness in balancing the pseudolabel and original hard label.

E. Impacts of Hyperparameters

As discussed in Section IV-B, five free parameters are empirically set for PERSVL, i.e., the number of blocks N_l in DBMI, the similarity threshold α, and the weights of different types of loss terms $\{\lambda_1, \lambda_2, \lambda_3\}$. In this section, we discuss their impacts by tuning their values. The values of N_l are increased from 1 to 4, the values of α are varied from 0.82 to 0.90 with an interval of 0.02, and the values of $\{\lambda_1, \lambda_2, \lambda_3\}$ are adjusted from 0 to 1 with an interval of 0.2. Note that only one parameter value could be changed at a time. For convenience, we only count the mR scores on the RSITMD dataset.

The performance of PERSVLs with different N_l is displayed in Fig. 5(a). We can find that the larger the N_l, the stronger the behavior of PERSVL. The PERSVLs' performance is enhanced dramatically when N_l increases from 1 to 3, and the growth tends to flatten out when N_l varies from 3 to 4. These observations indicate that the larger the DBMI, the more it contributes to the PERSVL. However, along with it comes an increase in the volume of parameters. Considering the trade-off between the DBMI's size and performance, we suggest $N_l = 3$. The results of PERSVLs with various α are shown in Fig. 5(b). It is easy to find that the performance of PERSVL shows an initial rise followed by a decline with α increases from 0.82 to 0.90, and the peak value appears at $\alpha = 0.86$. The reason is that when the filtered prior experiences are too few, the model can easily interfere with irrelevant prior experiences. Conversely, when the filtered prior experience is excessive, the model struggles to acquire adequate knowledge from prior experiences, making it challenging to correct noise during training effectively. Consequently, we suggest that the value of α could be fixed at 0.86.

The results of various PERSVLs with different λ_1–λ_3 are summarized in Fig. 5(c)–(e). It is easy to find that the performance of PERSVLs is enhanced steadily with three parameters increasing from 0 to 1. Taking λ_2 as an example, when its value equals 0, the mR score of PERSVL is as low as 40.32%. Once its value is raised to 1, PERSVL's mR score reaches 41.66%. These findings illustrate that different terms impact the overall loss function equally. In other words, the weights of different terms in $\mathcal{L}^{s1}$ and $\mathcal{L}^{s2}$ in (15) should be the same to each other.

F. Qualitative Analysis

1) Attention Map Visualization: To further validate the multimodal interaction capability of our PERSVL, we visualize the attention maps produced by DBMFE. Two RS images and their related texts are randomly selected from RSITMD, and the attention maps of each branch in DBMFE are exhibited separately in Fig. 6. From the results, we can observe that our PERSVL focuses on visual regions that are highly relevant to the textual semantics. Take the first image–text pair as an example. In the text branch, the model effectively focuses on the image regions related to "houses with blue roofs." In the image branch, the model not only accurately targets the described objects in the image (i.e., "house" and "blue roof") but also pays attention to the relationships between objects embedded in the image (such as "on" and "with").

2) Retrieval Examples: In this section, we display some retrieval examples. Three RS images and texts are selected randomly from three datasets as the queries. Then, their corresponding retrieval results are exhibited in Fig. 7. Due to space limitations, we only display the top-5 results. From the observation, we can see that our PERSVL achieves satisfactory performance, especially on the UCM-Captions dataset, our PERSVL accurately retrieves all five texts corresponding to the RS image in the image-to-text task. In addition, the model can retrieve other texts (RS images) that match the RS image (text) content but are labeled as 0. For example, in the RSICD dataset, PERSVL retrieves accurate descriptions like "many buildings and green trees are in an industrial area" and "many green plants with industrial beside in it" for the image-to-text task. All these results demonstrate the model's usefulness in RSITR.

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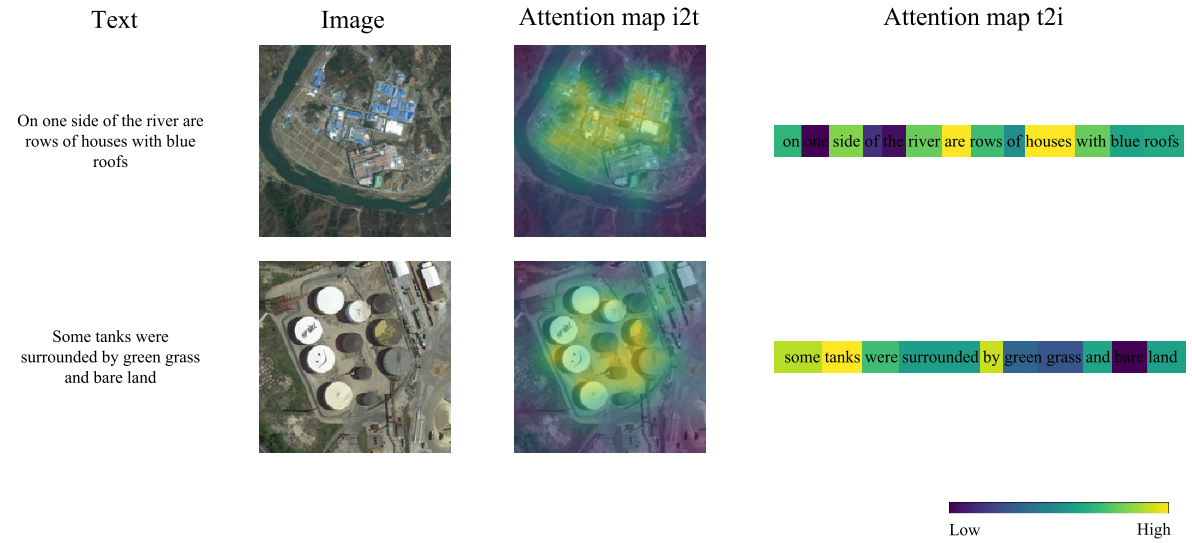

Fig. 6. Visualization of attention maps. Two RS image and text pairs are selected from RSITMD randomly. Their visual and textual attention maps generated by DBMFE are exhibited.

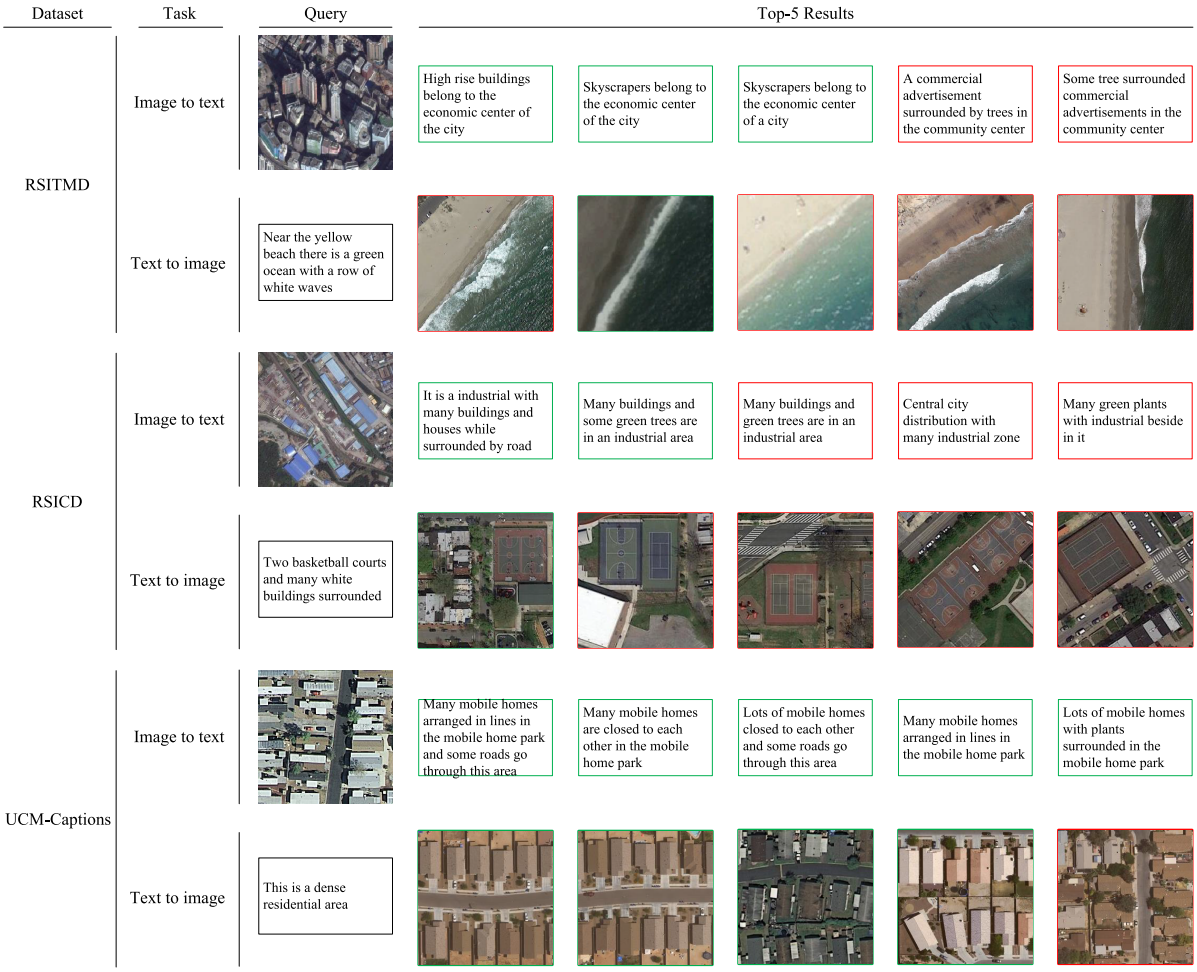

Fig. 7. Retrieval examples of PERSVL. We randomly select the three RS images and texts from three datasets as the queries to complete the "image-to-text" and "text-to-image" retrieval tasks. Due to space limitations, we only present the top-5 results. The correct/incorrect results are marked by green/red boxes.

V. Conclusion

Considering the complex contents within RS images and the intricate relations between RS images and texts, this article introduces a new model named PERSVL for RSITR tasks. It encloses four main components, i.e., a text encoder, a multiscale vision encoder, a DBMFE, and an LPE module. To address the current shortcomings of textual and visual encoders, we introduce BERT as the textual encoder and develop a multiscale vision encoder based on the Swin-Transformer. Combining them, the global connections within

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texts and multiscale clues hidden in RS images can be fully captured. Therefore, the data from different modalities can be mapped into embeddings effectively. To enhance the interaction between different embeddings, the DBMFE is proposed, which leverages the DBMI and branch fusion modules to accomplish cross-modal learning and produce matching scores for retrieval. Finally, the LPE module is introduced, which utilizes prior knowledge to generate reliable pseudolabels To mitigate the negative impacts caused by the FN sample problem.

Although our method achieves satisfactory RSITR results, there is still room for improvement. First, the parameter volume of the current version is relatively large. Lightweighting it while maintaining performance is one of our future works. Second, two encoders are used to learn the features independently, which may reduce the redundant information for cross-model learning. Overcoming this issue is another future work. We plan to introduce the large pretrained model with the prompt learning techniques for the first issue. In this way, not only can the texts and RS images be fully understood but also the number of parameters that need to be trained will also be limited. For the second issue, we consider guiding the text encoder to encode textual features from visual features and vice versa. This approach aims to extract textual (visual) features relevant to visual (textual) content and filter out irrelevant, redundant information. This can better facilitate cross-modal relationship learning between RS images and text.

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